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gcd(10, n) = 1

{ 2, 11, 111, ...  
has int many multiples of n  
→ p<sub>i</sub> = 1...1

→ gcd(10, n) = 1

↓

gcd(10, n):

a, b ∈ ℤ

a/10 + b/n = 1

a/b ∉ ℤ \ {0}

\* 1:      • b = 0,      a = 1      = 1 + 0(10)

• a = 0,      b = 1/n      1 = 0 + 1/n(10)

11: = 1 + 1(10)

111: = 12(10) + 1

(10 + 1)(10) + 1  
10<sup>2</sup> + 10 + 1

Σ<sub>i=0</sub><sup>2</sup> 10<sup>i</sup>

1111 = 1000 + 111 = 1000 + Σ<sub>i=0</sub><sup>2</sup> 10<sup>i</sup>  
= 10<sup>3</sup> + ...  
= Σ<sub>i=0</sub><sup>3</sup> 10<sup>i</sup>

p<sub>i</sub> = Σ<sub>j=1</sub><sup>i</sup> 10<sup>j-1</sup>

a/10 + b/n = 1  
10 + b/a n = 1/a  
b/a = 9      1/a = 10

$$p_i = 10 + r_i n$$

$$* a : \exists k \in \mathbb{Z} \quad ak = b$$

$$* \frac{10}{3} = \frac{9+1}{3} = 3 + \frac{1}{3}$$

$$\gcd(10, 3) = 1$$

$$\rightarrow 1 \mid 10 \quad \text{and} \quad 1 \mid 3$$

$$\frac{10}{1} = a \quad \frac{3}{1} = b$$

$$10 = a \cdot 1 \quad \text{and} \quad 3 = b \cdot 1$$

$$13 = a \cdot 1 + b \cdot 1$$

$$13 = (a+b) \cdot 1$$

$$1 = \frac{1}{(a+b)} \cdot 10 + \frac{1}{(a+b)} \cdot 3$$

$$1 = c \cdot 10 + d \cdot 3$$

$$1 = x \cdot 10 + y \cdot 3$$

$$* \boxed{a \cdot 10 + b \cdot n = 1}$$

$$I, II, III, IV, \dots$$

$$p_i =$$

$$n: (a, b)$$

$$10 + \frac{b}{a} n = \frac{1}{a}$$

$$10 + r n = l$$

$$\rightarrow \text{Evel. : } x, y : \exists q, l \text{ s.t. } x = qy + l$$

Relate  $n$  and  $p_i$  for any  $p_i$

$$* \boxed{p_i = q_i n + l_i} *$$

$$p_i \in \mathbb{Z}$$

$$* p_i \text{ also satisfies } p_{i+1} =$$

Counting Problem:

$$X \xrightarrow{f} \mathbb{N}$$

$$X = \{p_i\}_{i=1}$$

$$f : X \rightarrow \mathbb{N}$$

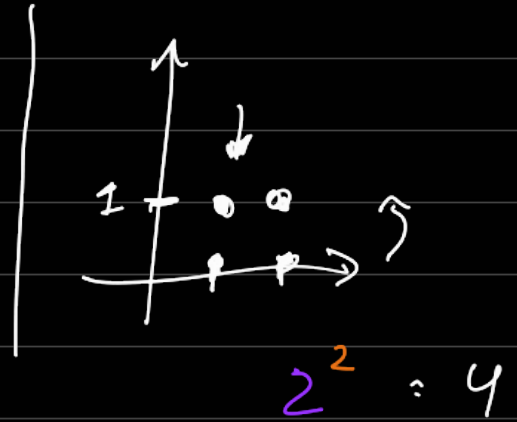
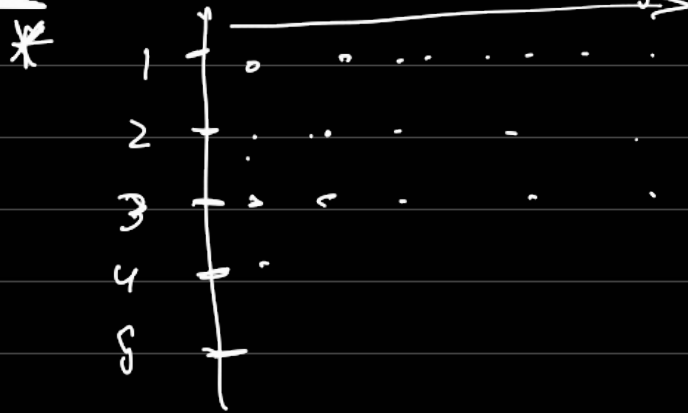
$$l_i = 0, 1, 2, \dots, n-1$$

$$f(p_i) = l_i$$

$$q \in \mathbb{Q}$$

$$\hookrightarrow f(p_i) = i$$

Show is that  $\text{inj.}$  and  $\text{sur.}$



\* 3 points:

$$5 \text{ pts: } 2^5 = 32$$

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$\leadsto$   $\mathbb{N}$ : infinite points

$$x = i, \text{ or } j \quad x \in \{1, \dots, 5\}$$

$$m \times n \text{ or } [m] \times [n]$$

• Idea: After going past the  $[5] \times [5]$  section  
 $[5] \times [m]$

$$f: [5] \times [d] \longrightarrow S$$

$S$ : satisfy P.H. princ.

\* Math:

\* "In words"

$n$ : items

$\hookrightarrow$  fall into  $m$  categories  
 $n > m$

then two of the same items  
 must fall into the same category

$n$ : color num

$m$ : 5

$$S: \{1, 2, 3, 4, 5\}$$

$$d > 32$$

$$d = 3$$

Codomain      Domain  
 $\downarrow$        $\downarrow$   
 Grid      Pattern

$$f: [5] \times [33] \xrightarrow{32} 5$$

Codomain      Domain

$$\hookrightarrow d = 2^n + 1 \quad \text{General}$$

20)  $\{H_i\}_{i \in \mathbb{Z}}$

(a)  $\bigcap H_i$

(i) identity:  $\forall i, e \in H_i \Rightarrow e \in \bigcap H_i$

(ii) Closure:  $x, y \in H \Rightarrow x \cdot y \in H$   
 $x, y \in \bigcap H_i$

then  $\exists \forall n$  s.t.  $x \in H_n$  and  $y \in H_n$

then  $x \cdot y \in H_n \Rightarrow \in \bigcap H_i$

(b)  $\langle X \rangle = \bigcap_{\substack{H \leq G \\ X \subset H}} H$

$$\langle X \rangle = \{ \quad \quad \quad \}$$

(i) Id.

(ii) closure

$$X = \{x_1, x_2, \dots\}$$

(iii) inverse

$\hookrightarrow$  Free Group

$$F(X) = \{ x_1^{\alpha_1} x_2^{\alpha_2} \dots \mid \}$$

$$X = \{x_1, x_2, \dots\}$$

$$X \subset G$$

$$X \subset H_i$$

$$\bullet X \subset \langle X \rangle \bullet$$

(i)  $e \in \langle X \rangle$

(ii)  $x_i * x_i \dots$

100 times

$$\in \langle X \rangle$$

$$x_i \cdot x_i \text{ etc. } \dots$$

• Any product of elements of  $X \in \langle X \rangle$

(iii) Any product of inv. " "  $\in \langle x \rangle$

$$\langle x \rangle = \{ \underbrace{x^{e_1} \dots x^{e_n}}_{n \geq 0} \mid e_n \in \{+1, -1\} \}$$

③  $f: \mathbb{Z} \rightarrow G$

a group homomorphism looks like:

$$f(a \cdot b) = f(a) * f(b)$$

•  $(\mathbb{Z}, +)$

$$f(a+b) = f(a) * f(b)$$

$$f(0) = e$$

$(\mathbb{Z})$ : "built" from 1

$f(1) = g \in G$   
 $\hookrightarrow n \in \mathbb{Z}$

$$f(-1) = g^{-1} \quad g^{-1} = "1"$$

$$\begin{aligned} f(n) &= f(1+1+\dots+1) \\ &= f(1) * f(1) * \dots * f(1) \\ &= g * g * \dots * \\ &= g^n \end{aligned}$$

$$f(-n) = g^{-n}$$

$$\text{Im}(f) = \{ g^n \mid n \in \mathbb{Z} \} = \underbrace{\langle g \rangle}$$

④  $\langle g \rangle \cong \mathbb{Z}/n\mathbb{Z}$

